SIMATS SCHOOL OF ENGINEERING

Approved By AICTE | IET-UK Accreditation

Course Code: DSA0311	Course Name: Natural Language Processing
Branch: AI&DS	Year : 2024-2028
Name	A.Abhishek Reddy
Rg no	192424192

Assignment 4

1. Give a FOL translations for the following sentences: (10M) (CO4) (BLT4)

a. Vegetarians do not eat meat.

b. Not all vegetarians eat eggs.

2. Give a set of facts and inferences necessary to prove the following assertions: (10M) (CO4) (BLT4)

a. McDonalds is not a vegetarian restaurant.

b. Some vegetarians can eat at McDonalds.

Don't just place these facts in your knowledge base. Show that they can be inferred from some more general facts about vegetarians and McDonalds.

3. Give FOL translations for the following sentences that capture the temporal relationships between the events. (10M) (CO4) (BLT4)

a. When Mary's flight departed, I ate lunch.

b. When Mary's flight departed, I had eaten lunch.

4. The following rule, which we gave as a translation, is not a reasonable definition of what it

means to be a vegetarian restaurant. (10M) (CO4) (BLT4)

$\forall x \text{VegetarianRestaurant}(x) \Rightarrow \text{Serves}(x,\text{VegetarianFood})$

QUESTION 1 – FOL TRANSLATIONS

1. Introduction to First-Order Logic

First-Order Logic (FOL) is a formal system used to represent facts, objects, relationships, and rules. It is widely used in Artificial Intelligence and Natural Language Processing to represent knowledge in a structured form.

FOL uses predicates, variables, constants, logical operators, and quantifiers to represent sentences.

Main components of FOL

1. Constants: Represent specific objects or individuals. Example: Mary, John, McDonalds.
2. Variables: Represent general objects. Example: x, y, z.
3. Predicates: Describe properties or relationships. Example: Vegetarian(x), Eats(x,y).
4. Quantifiers: Represent the quantity of objects.
5. Logical operators: Connect different logical statements.

Logical symbols

Symbol	Meaning
∀	For all
∃	There exists
¬	Not
∧	AND
∨	OR
→	Implies
↔	If and only if

2. Question 1(a): Vegetarians do not eat meat.

Explanation

This sentence states that every person who is a vegetarian does not eat meat. It expresses a general rule that applies to all vegetarians.

Predicates

- Vegetarian(x): x is a vegetarian.
- Eats(x,y): x eats y.
- Meat(y): y is meat.

FOL Translation

\[\forall x\forall y\, [(Vegetarian(x)\land Meat(y))\rightarrow \neg Eats(x,y)] \]

Explanation of the formula

1. ∀x means for every person.

2. ∀y means for every food item.
3. Vegetarian(x) identifies a vegetarian person.
4. Meat(y) identifies a meat food item.
5. ¬Eats(x,y) means the person does not eat meat.
6. The implication represents the general rule.

Example

If Ravi is a vegetarian and chicken is meat, the rule allows us to conclude that Ravi does not eat chicken.

$$ Vegetarian(Ravi) $$

$$ Meat(Chicken) $$

Therefore,

$$ \neg Eats(Ravi,Chicken) $$

3. Question 1(b): Not all vegetarians eat eggs.

Explanation

This sentence means that at least one vegetarian does not eat eggs. It does not mean that every vegetarian avoids eggs.

Predicates

- Vegetarian(x): x is a vegetarian.
- Eats(x,y): x eats y.
- Egg(y): y is an egg.

FOL Translation

$$ \exists x\exists y\, [Vegetarian(x)\land Egg(y)\land \neg Eats(x,y)] $$

Explanation of the formula

1. ∃x means there exists at least one person.
2. ∃y means there exists an egg.
3. Vegetarian(x) means the person is vegetarian.
4. Egg(y) means the food item is an egg.
5. ¬Eats(x,y) means the person does not eat that egg.

Alternative logical form

$$ \neg\forall x[Vegetarian(x)\rightarrow Eats(x,Eggs)] $$

This means it is not true that every vegetarian eats eggs.

Example

Suppose Ravi and Kumar are vegetarians. Ravi eats eggs, but Kumar does not eat eggs. This example satisfies the statement because at least one vegetarian avoids eggs.

4. Difference Between Question 1(a) and 1(b)

Question	Meaning	Quantifier
Vegetarians do not eat meat	Every vegetarian avoids meat	Universal (∀)
Not all vegetarians eat eggs	At least one vegetarian does not eat eggs	Existential (∃)

5. Applications of FOL

1. Represents knowledge in Artificial Intelligence.
2. Helps computers understand logical statements.
3. Supports automated reasoning.
4. Represents relationships between objects.
5. Helps develop expert systems.

6. Conclusion

First-Order Logic converts natural language sentences into mathematical expressions. The two statements demonstrate how universal and existential quantifiers represent different meanings. FOL makes knowledge representation clear and supports logical reasoning in AI systems.

QUESTION 2 – FACTS AND INFERENCES

1. Introduction

Knowledge representation is the process of storing information in a form that a computer can understand and use for reasoning. In First-Order Logic, facts and rules are combined to derive new conclusions.

Inference is the process of using known facts and general rules to prove a statement.

The given problem requires proving:

a. McDonalds is not a vegetarian restaurant.

b. Some vegetarians can eat at McDonalds.

These conclusions must be derived from general facts about restaurants, food, and vegetarians rather than directly entered as facts.

2. Basic Predicates

The following predicates are used:

Predicate	Meaning
Restaurant(x)	x is a restaurant
VegetarianRestaurant(x)	x is a vegetarian restaurant
Vegetarian(x)	x is a vegetarian person
Serves(x,y)	Restaurant x serves food y
Meat(y)	y is meat
Eats(x,y)	Person x eats food y
CanEatAt(x,y)	Person x can eat at restaurant y

Constants:

- McDonalds: A particular restaurant.
- MeatFood: A meat food item.
- VegPerson: A vegetarian person.

3. General Facts and Rules

The following general rules form the knowledge base.

Rule 1: A vegetarian restaurant does not serve meat.

[forall r\forall f[(VegetarianRestaurant(r)\land Meat(f)) \rightarrow \neg Serves(r,f)]]

Rule 2: A restaurant that serves meat is not a vegetarian restaurant.

[forall r\forall f[(Restaurant(r) land Meat(f)land Serves(r,f)) rightarrow neg VegetarianRestaurant(r)] \]

Rule 3: A vegetarian can eat food that does not contain meat.

[\forall p\forall r[(Vegetarian(p)\land Restaurant(r)\land SuitableFoodAvailable(r,p)) \rightarrow CanEatAt(p,r)] \]

Rule 4: A restaurant serves different types of food.

\[\forall r[Restaurant(r)\rightarrow OffersFood(r)] \]

4. Specific Facts About McDonalds

The following facts describe the restaurant and its menu.

1. McDonalds is a restaurant.

\[Restaurant(McDonalds) \]

2. McDonalds serves at least one meat item.

\[Meat(Burger) \]

$$ Serves(McDonalds,Burger) $$

3. McDonalds also serves a vegetarian food item.

$$ VegetarianFood(VegBurger) $$

$$ Serves(McDonalds,VegBurger) $$

4. A vegetarian can eat the vegetarian food item.

$$ SuitableFoodAvailable(McDonalds,VegetarianCustomer) $$

5. At least one vegetarian person exists.

$$ Vegetarian(Ravi) $$

5. Inference for Assertion (a)

Assertion: McDonalds is not a vegetarian restaurant.

Step 1: Identify the facts

$$ Restaurant(McDonalds) $$

$$ Meat(Burger) $$

$$ Serves(McDonalds,Burger) $$

Step 2: Apply Rule 2

$$ (Restaurant(r)\land Meat(f)\land Serves(r,f)) \rightarrow \neg VegetarianRestaurant(r) $$

Step 3: Substitute the values

Let:

- r = McDonalds
- f = Burger

Therefore,

$$ \begin{aligned} &Restaurant(McDonalds)\\ &\land Meat(Burger)\\ &\land Serves(McDonalds,Burger) \end{aligned} $$

Step 4: Apply inference

Since all the conditions are true, the conclusion is:

$$ \boxed{\neg VegetarianRestaurant(McDonalds)} $$

Assertion (a) proved

McDonalds is not classified as a vegetarian restaurant because it serves meat.

6. Inference for Assertion (b)

Assertion: Some vegetarians can eat at McDonalds.

Step 1: Identify the facts

$$ Vegetarian(Ravi) $$

$$ Restaurant(McDonalds) $$

$$ SuitableFoodAvailable(McDonalds,Ravi) $$

Step 2: Apply Rule 3

$$ (Vegetarian(p)\land Restaurant(r)\land SuitableFoodAvailable(r,p)) \rightarrow CanEatAt(p,r) $$

Step 3: Substitute the values

Let:

- p = Ravi
- r = McDonalds

Therefore,

$$ \begin{aligned} &Vegetarian(Ravi)\\ &\land Restaurant(McDonalds)\\ &\land SuitableFoodAvailable(McDonalds,Ravi) \end{aligned} $$

Step 4: Apply inference

$$ CanEatAt(Ravi,McDonalds) $$

Since Ravi is one vegetarian person who can eat at McDonalds, we conclude:

$$ \boxed{\exists x[Vegetarian(x)\land CanEatAt(x,McDonalds)]} $$

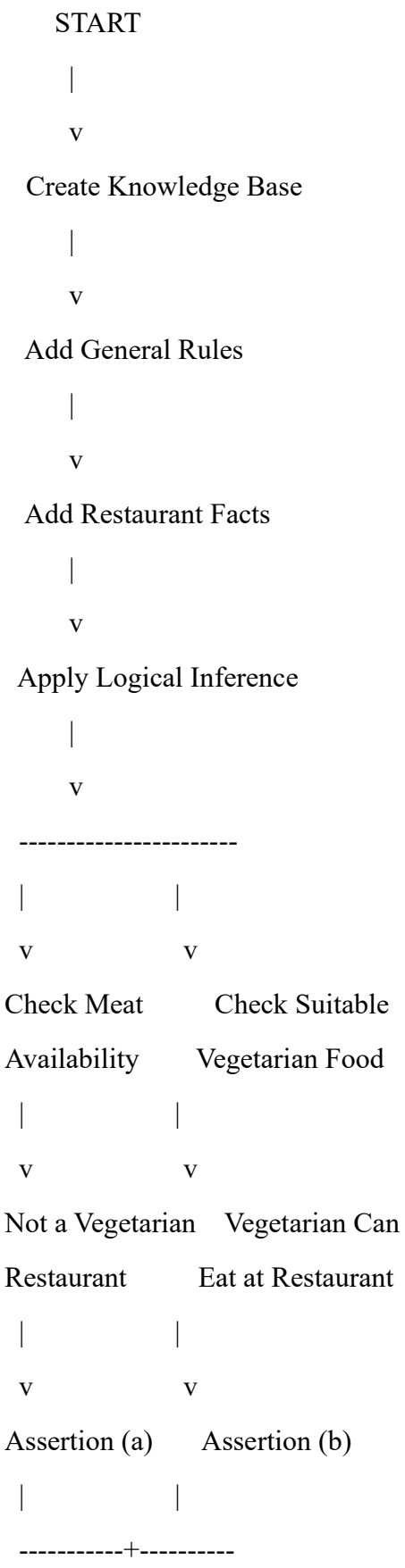
Assertion (b) proved

Some vegetarians can eat at McDonalds because vegetarian food options are available.

7. Knowledge Base and Inference Table

Type	**Statement**
General rule	Vegetarian restaurants do not serve meat
General rule	Restaurants serving meat are not vegetarian restaurants
General rule	Vegetarians can eat at restaurants with suitable food
Fact	McDonalds is a restaurant
Fact	McDonalds serves a meat burger
Fact	McDonalds has a suitable vegetarian food option
Inference 1	McDonalds is not a vegetarian restaurant
Inference 2	Some vegetarians can eat at McDonalds

8. Flowchart of Inference



|

v

END

9. Applications

1. Used in expert systems.
2. Helps in automated decision-making.
3. Represents real-world knowledge.
4. Supports logical problem-solving.
5. Used in artificial intelligence applications.

10. Conclusion

The knowledge base uses general rules and specific facts to prove both assertions. McDonalds is not a vegetarian restaurant because it serves meat. However, some vegetarians can eat there because suitable vegetarian food is available.

QUESTION 3 – FOL TRANSLATIONS FOR TEMPORAL RELATIONSHIPS

1. Introduction

Temporal relationships describe when an event happens in relation to another event. In Natural Language Processing, temporal logic helps computers understand the order and timing of events in a sentence.

First-Order Logic can represent events using predicates and variables. Time information can be added to show whether an event happened before, during, or after another event.

The given sentences describe the relationship between two events:

- Mary's flight departed.
- I ate lunch.

The difference between the two sentences is the time at which lunch was eaten.

2. Basic Predicates

The following predicates are used:

Predicate	**Meaning**
FlightDeparted(x,t)	Event x departed at time t
AteLunch(p,t)	Person p ate lunch at time t
Before(t1,t2)	Time t1 is before time t2

After(t1,t2)	Time t1 is after time t2
Person(p)	p is a person

Constants:

- MaryFlight: Mary's flight.
- I: The speaker.
- LunchEvent: The lunch event.

3. Question 3(a): When Mary's flight departed, I ate lunch.

Explanation

This sentence indicates that the speaker ate lunch at the same time Mary's flight departed.

Both events happen at the same time.

FOL Translation

[exists t[FlightDeparted(MaryFlight,t) \land AteLunch(I,t)]]

Explanation of the formula

1. ∃t means there exists a time.
2. FlightDeparted(MaryFlight,t) means Mary's flight departed at time t.
3. AteLunch(I,t) means the speaker ate lunch at the same time.
4. ∧ connects both events.

Example

Suppose Mary's flight departed at 1:00 PM. The speaker also ate lunch at 1:00 PM.

Event	**Time**
Mary's flight departed	1:00 PM
I ate lunch	1:00 PM

Both events occur at the same time

Timeline

Time ---------->

1:00 PM

|

----------+----------

| |

Flight departed Ate lunch

(Mary) (I)

4. Question 3(b): When Mary's flight departed, I had eaten lunch.

Explanation

This sentence indicates that the speaker completed eating lunch before Mary's flight departed.

The phrase “had eaten” represents an action completed before another past event.

FOL Translation

\[\exists t_1\exists t_2[AteLunch(I,t_1) \land FlightDeparted(MaryFlight,t_2) \land Before(t_1,t_2)] \]

Explanation of the formula

1. $\exists t_1$ represents the time when the speaker ate lunch.
2. $\exists t_2$ represents the time when Mary's flight departed.
3. AteLunch(I,t_1) means the speaker ate lunch at time t_1.
4. FlightDeparted(MaryFlight,t_2) means the flight departed at time t_2.
5. Before(t_1,t_2) means lunch happened before the flight departed.

Example

Suppose the speaker completed lunch at 12:00 PM and Mary's flight departed at 1:00 PM.

Event	**Time**
I ate lunch	12:00 PM
Mary's flight departed	1:00 PM

Lunch happened before the flight departed

Timeline

Time ---------->

12:00 PM 1:00 PM

| |

v v

Ate lunch Flight departed

(I) (Mary)

5. Difference Between Both Sentences

Feature	Sentence (a)	Sentence (b)
Sentence	I ate lunch	I had eaten lunch
Meaning	Both events happen at the same time	Lunch happened earlier
Time relationship	Simultaneous	Before
FOL condition	Same time variable	Before(t_1,t_2)

6. Importance of Temporal Relationships

1. Helps identify the order of events.
2. Helps understand past and present actions.
3. Supports sentence interpretation.
4. Helps computers understand time-based information.
5. Improves natural language understanding.

7. Applications

1. Used in chatbots to understand event timing.
2. Used in question-answering systems.
3. Helps summarize stories and documents.
4. Supports event extraction from text.
5. Used in intelligent information retrieval.

8. Conclusion

Temporal relationships help identify when events occur. In the first sentence, lunch and the flight departure happen at the same time. In the second sentence, lunch is completed before the flight departs. FOL represents these differences using time variables and the Before relationship.

QUESTION 4 – ANALYSIS OF THE FOL RULE FOR A VEGETARIAN RESTAURANT

1. Introduction

First-Order Logic (FOL) is used to represent facts and rules in Artificial Intelligence. A logical rule must correctly describe the meaning of a sentence.

The given rule attempts to define a vegetarian restaurant. However, it is not a reasonable definition because it does not completely describe what a vegetarian restaurant should provide.

2. Given Rule

\[\forall x\,[VegetarianRestaurant(x) \Rightarrow Serves(x,VegetarianFood)] \]

Meaning: Every vegetarian restaurant serves vegetarian food.

3. Explanation of the Rule

- ∀x means for every restaurant.
- VegetarianRestaurant(x) means x is a vegetarian restaurant.
- Serves(x, VegetarianFood) means the restaurant serves vegetarian food.
- ⇒ means implies.

The rule states that if a restaurant is vegetarian, it must serve vegetarian food.

4. Why Is This Rule Not Reasonable?

The given rule is incomplete because:

1. It only states that a vegetarian restaurant serves vegetarian food.
2. It does not say that the restaurant must avoid serving meat.
3. A restaurant may serve both vegetarian and non-vegetarian food.
4. Serving vegetarian food alone does not make a restaurant completely vegetarian.
5. Therefore, the rule cannot correctly distinguish vegetarian restaurants from other restaurants.

5. Example

Consider two restaurants:

Restaurant	Vegetarian Food	Meat Food
Restaurant A	Yes	No
Restaurant B	Yes	Yes

According to the given rule, both restaurants satisfy the condition because both serve vegetarian food.

However, Restaurant B also serves meat. Therefore, it should not be classified as a vegetarian restaurant.

This shows that the given rule is not sufficient.

6. Corrected FOL Rule

A better definition should state that a vegetarian restaurant serves vegetarian food and does not serve meat.

\[\forall x[VegetarianRestaurant(x) \leftrightarrow (Serves(x,VegetarianFood) \land \neg Serves(x,MeatFood))] \]

7. Explanation of the Corrected Rule

- VegetarianRestaurant(x) means x is a vegetarian restaurant.
- Serves(x, VegetarianFood) means it serves vegetarian food.
- ¬Serves(x, MeatFood) means it does not serve meat.
- ∧ means both conditions must be true.
- ↔ means both sides imply each other.

Therefore, a restaurant is considered vegetarian only when it serves vegetarian food and does not serve meat.

8. Comparison of the Rules

Given Rule	**Corrected Rule**
Only checks vegetarian food	Checks vegetarian food and absence of meat
Uses implication (⇒)	Uses equivalence (↔)
Does not prevent serving meat	Prevents serving meat
Incomplete definition	More complete definition

9. Flowchart

START

|

v

Check Restaurant

|

v

Does it serve vegetarian food?

|

Yes | No

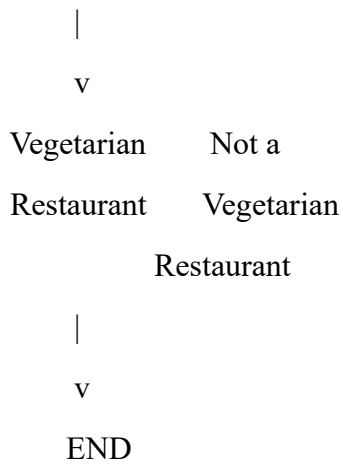
|

v

Does it serve meat?

|

No | Yes



10. Applications

1. Used in knowledge representation.
2. Helps classify restaurants.
3. Supports logical decision-making.
4. Helps identify incomplete rules.
5. Improves AI reasoning systems.

11. Conclusion

The given rule is not a reasonable definition because it only checks whether a restaurant serves vegetarian food. It does not prevent the restaurant from serving meat. The corrected rule includes both conditions: serving vegetarian food and not serving meat. Therefore, it provides a more complete definition of a vegetarian restaurant.